Convolutional Neural Networks (CNN)

Cathy Jimenez

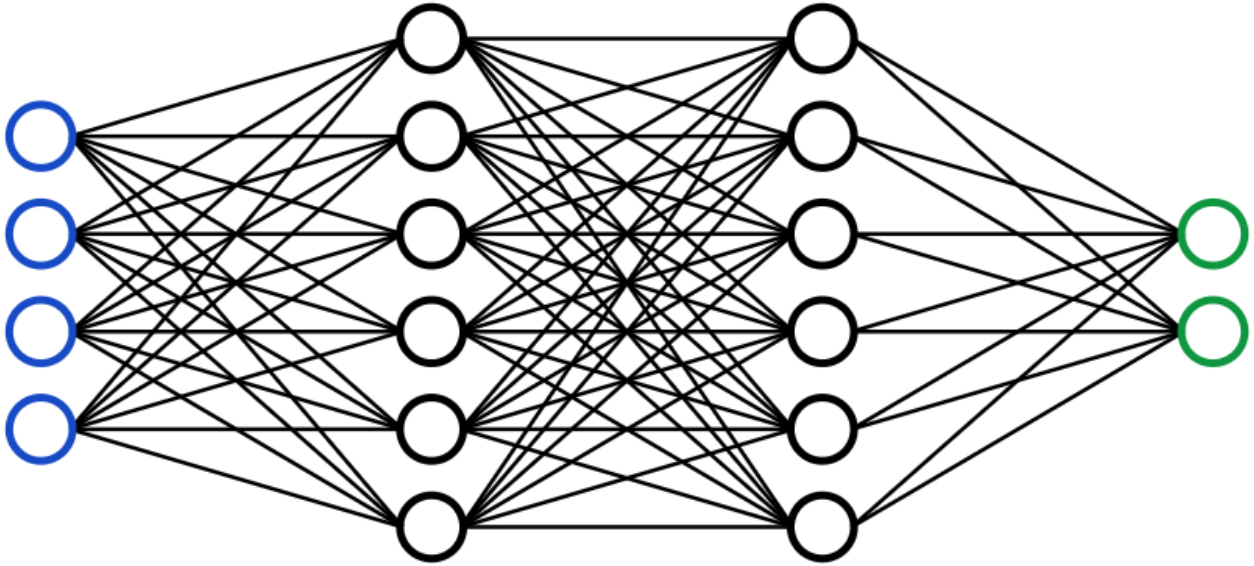
Data Analysis II

From Pixels to Predictions, An Intuitive Approach

- The Problem
- The Biological Inspiration
- End-to-End CNN Workflow
- Technical Demo
- Future Implementations

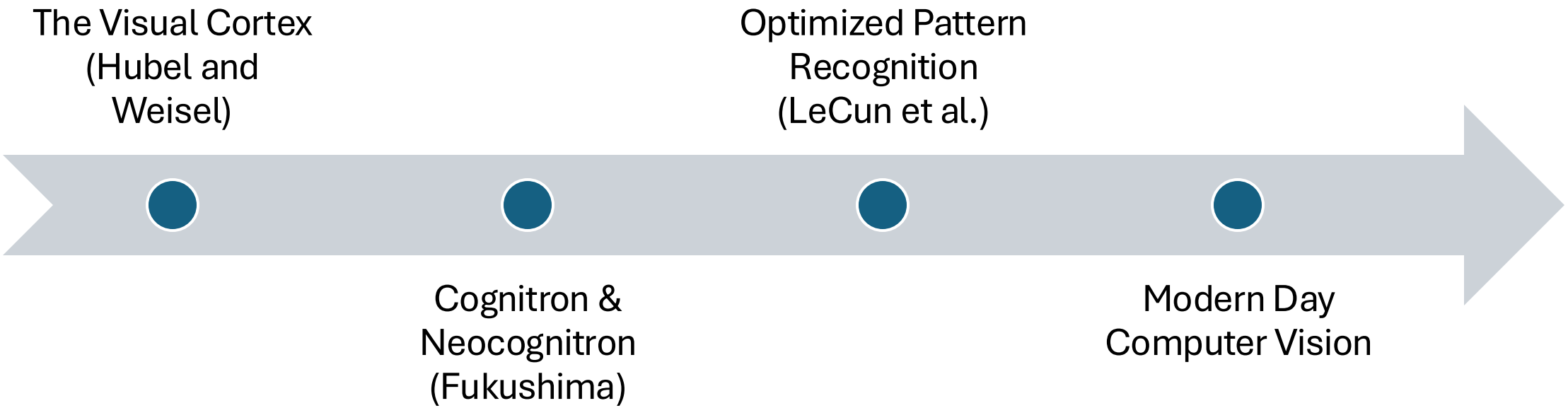
Identifying the Problem In CNNs

- Computers process images as matrices of numbers
- Spatial relationships between pixels
- Massive blowup in the number of network parameters



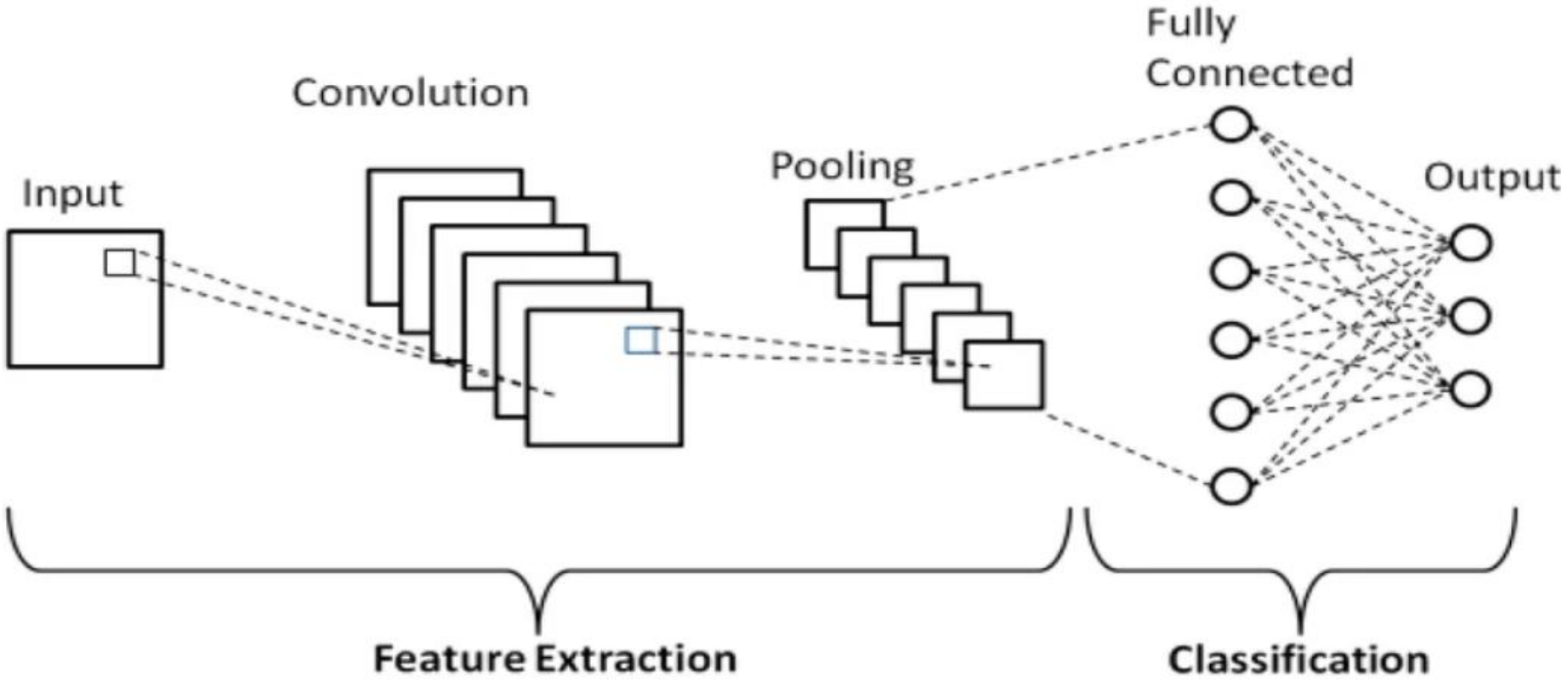
CNNs Emulate Biological Findings

- Visual patterns
- Local receptive fields and shared weights



The End-to-End CNN Workflow

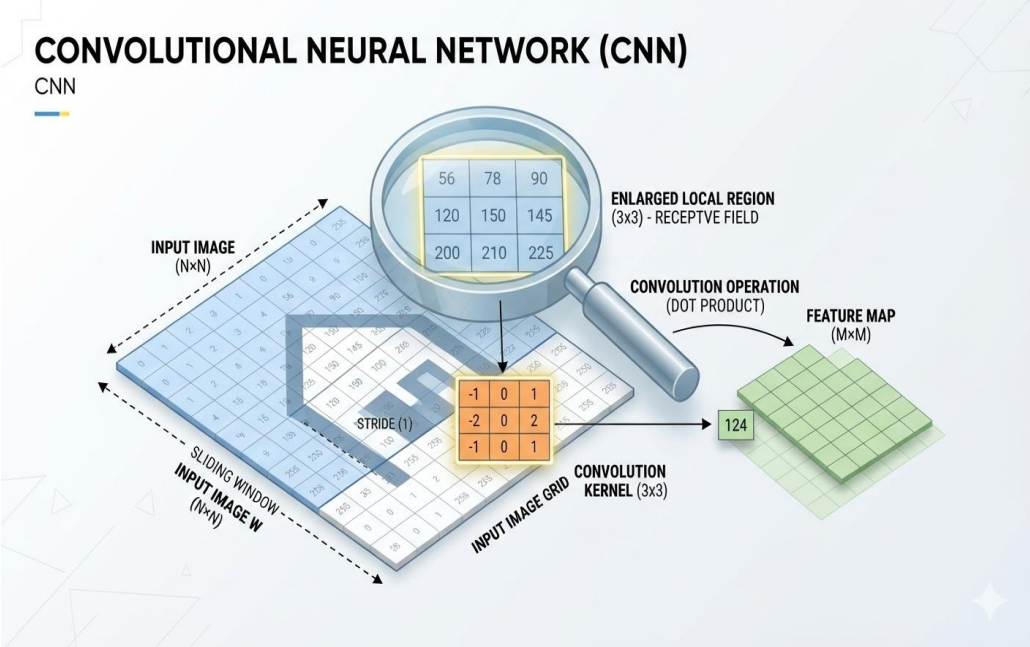




Source: Phung, Van & Rhee, Eun. (2019). A High-Accuracy Model Average Ensemble of Convolutional Neural Networks for Classification of Cloud Image Patches on Small Datasets. Applied Sciences. 9.4500.10.3390/app9214500.

Convolution: Divide and Conquer

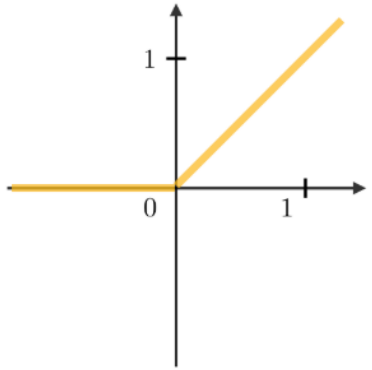
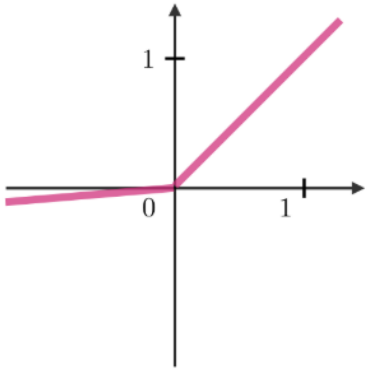
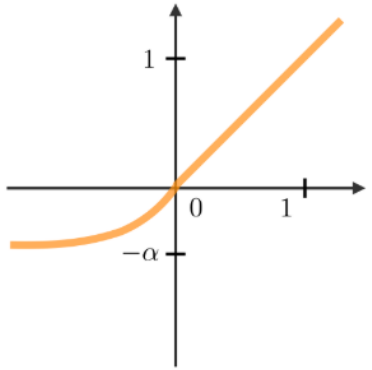
- Local Receptive Fields
- The Filtering Process, *"The Magnifier Glass"*
- Feature Maps and visual features
- Convolutions, mapping local connectivity and learning distinct spatial patterns



Note. Image generated using the prompt "put a magnifying glass over the input," by Gemini, 2026 (google.com).

Non-Linearity for Real-World Data

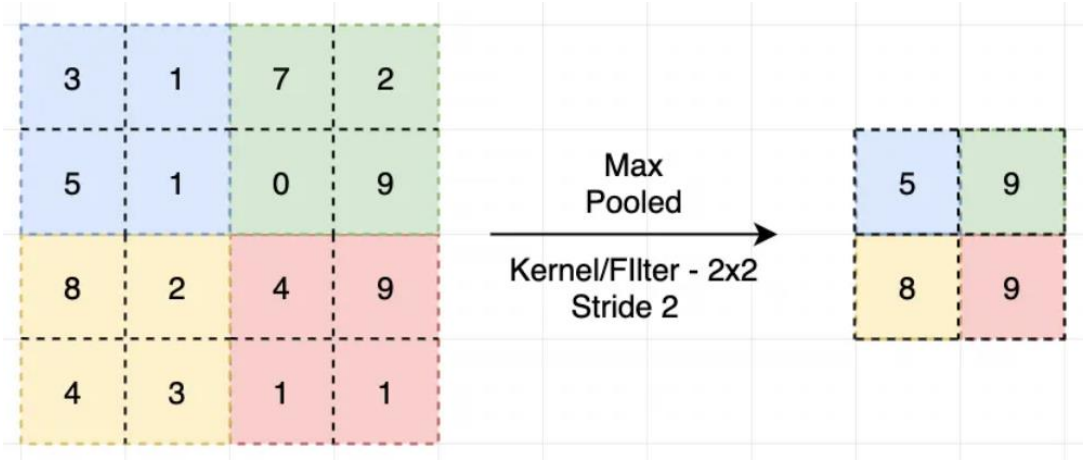
- Feature Map transformations to add complexity

ReLU	Leaky ReLU	ELU
$g(z) = \max(0, z)$	$g(z) = \max(\epsilon z, z)$ with $\epsilon \ll 1$	$g(z) = \max(\alpha(e^z - 1), z)$ with $\alpha \ll 1$
		
• Non-linearity complexities biologically interpretable	• Addresses dying ReLU issue for negative values	• Differentiable everywhere

Amidi, A., & Amidi, S. (n.d.). CS 230 - Convolutional Neural Networks cheatsheet. Stanford University. https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks/#activation

Different Perspectives: Pooling / Downsampling

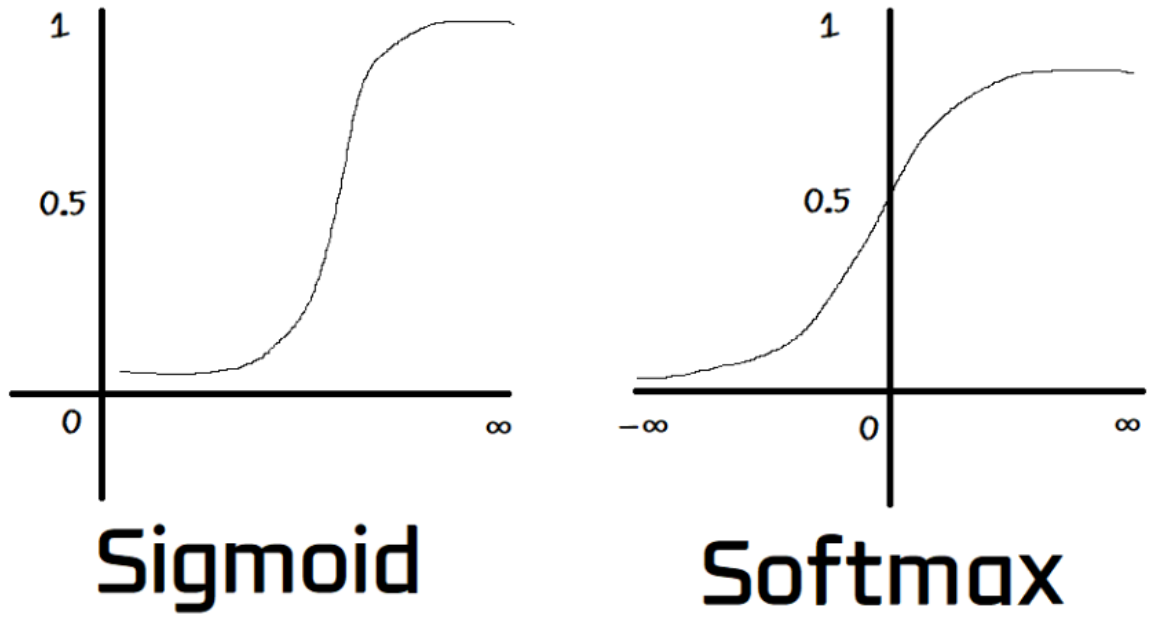
- Reduces spatial resolution, typically via Max Pooling
- Deeper layers used to scan broader, more abstract configurations
- From Simple lines and edges to complex shapes and objects
- *"The Magnifying Glass"*



Rana, K. (2020, April 20). *Pooling layer — Short and simple*. In Plain English. https://plainenglish.io/artificial-intelligence/pooling-layer-beginner-to-intermediate-fa0dbdce80eb

The Learning Engine & The Loss Function

- The Dense Layers
- Optimization
  - Weight changes
  - Loss function
- Probability Distribution
- Final Output



Chiro-J. (2022, August 9). #13 Softmax Function. Velog. velog.io

Technical Implementation Demo

- Training dataset of 60,000 images
- 28x28 grayscale images
- 10 digits
- Test set of 10,000 images
- Pixel values range from 0-255

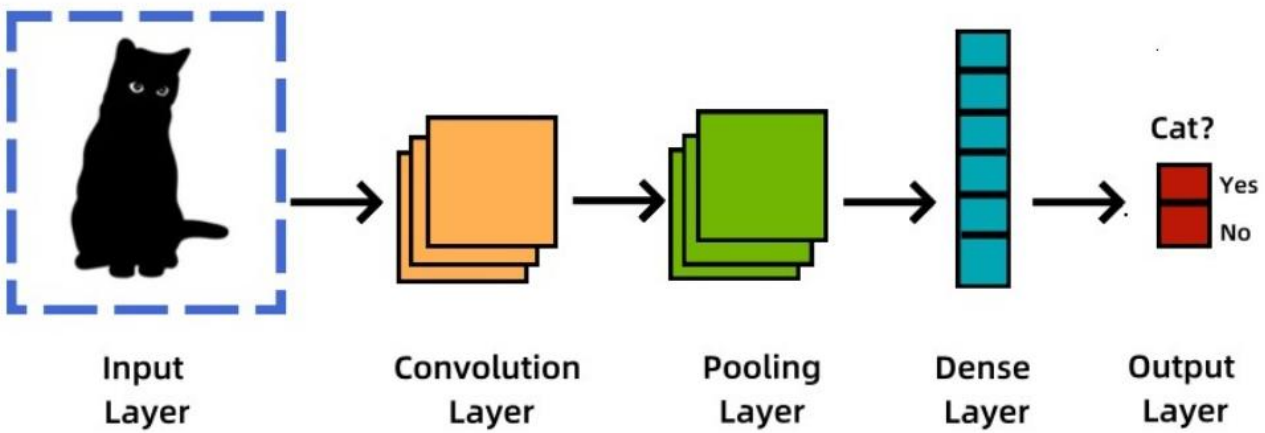
Future Implementations

- **Project Overview & Dataset**
  - **Multi-class classification** utilizing Convolutional Neural Network (CNN) modeling.
  - **Data Source: COVID-19 Radiography Database** (Kaggle), compiled by Qatar University, University of Dhaka, and international partners.
  - **Dataset Composition: 21,165 total Chest X-Ray (CXR) images** across four distinct classes:
    - **Normal / Healthy:** 10,192 images
    - **Lung Opacity (Non-COVID):** 6,012 images
    - **COVID-19 Positive:** 3,616 images
    - **Viral Pneumonia:** 1,345 images

Methodological Extensions

- Can the model serve as a reliable second opinion?
- How close are the COVID-19 and viral pneumonia predictions?

Questions?


Cat?
Yes
No
Input Layer
Convolution Layer
Pooling Layer
Dense Layer
Output Layer

References

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- LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278–2324. https://doi.org